Maximum Likelihood Estimation

Where every loss function comes from

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Mathematical Foundations for AI · IIT Gandhinagar

Deriving losses from probability models

Three lines you will type for the rest of your career

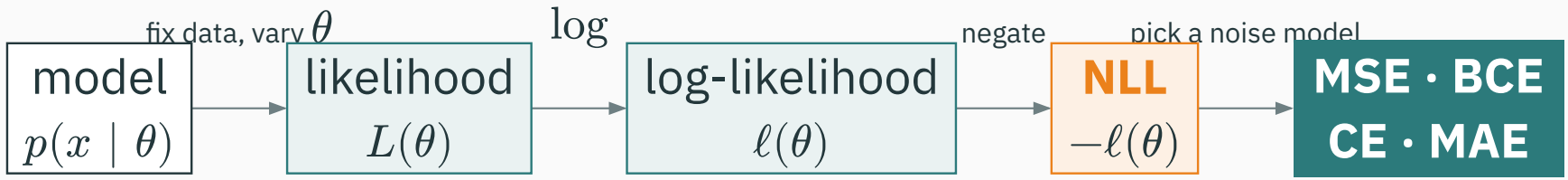
Every ML training loop — every paper, every framework — ends the same way:

```
loss = F.mse_loss(y_hat, y)              # regression
loss = F.binary_cross_entropy(p, y)      # yes/no classification
loss = F.cross_entropy(logits, y)        # multi-class classification
```

Squaring errors. Taking logs of probabilities. **Who decided these formulas?** Why squared — not absolute, not fourth power? Why is there a logarithm inside a classification loss?

Nobody tuned these by trial and error. All three fall out of **one idea**, mechanically. Today: the idea, and the full derivation for MSE. Cross-entropy's turn comes in L24 — but you'll leave today knowing exactly where it lives.

From a probability model to an objective



Fix the observed data and vary the parameters. The following slides compute every step in this map.

You have never formally "fit a model" in this course. That is the point: today is where **fitting** becomes a mathematical operation instead of a vibe.

Learning outcomes

By the end of this lecture you will be able to:

1. Read $p(x \mid \theta)$ **two ways**: as a distribution over x, and as a likelihood over θ.
2. Write the likelihood of an entire dataset using the **IID assumption**.
3. Explain **three independent reasons** the log goes in — and recap L2's 0.03^{1000} disaster.
4. Derive $\hat{\theta}_{\text{MLE}}$ for the **Bernoulli** coin (⭐) and the **Gaussian** mean.
5. Derive the course's keystone: **minimizing MSE = maximizing Gaussian likelihood** (⭐).
6. Map noise models to the losses they generate: Gaussian→MSE, Bernoulli→BCE, Categorical→CE, Laplace→MAE.

The flip: from probability to likelihood

Pop quiz · one student, three courses

Three courses grade on curves. Their score distributions:

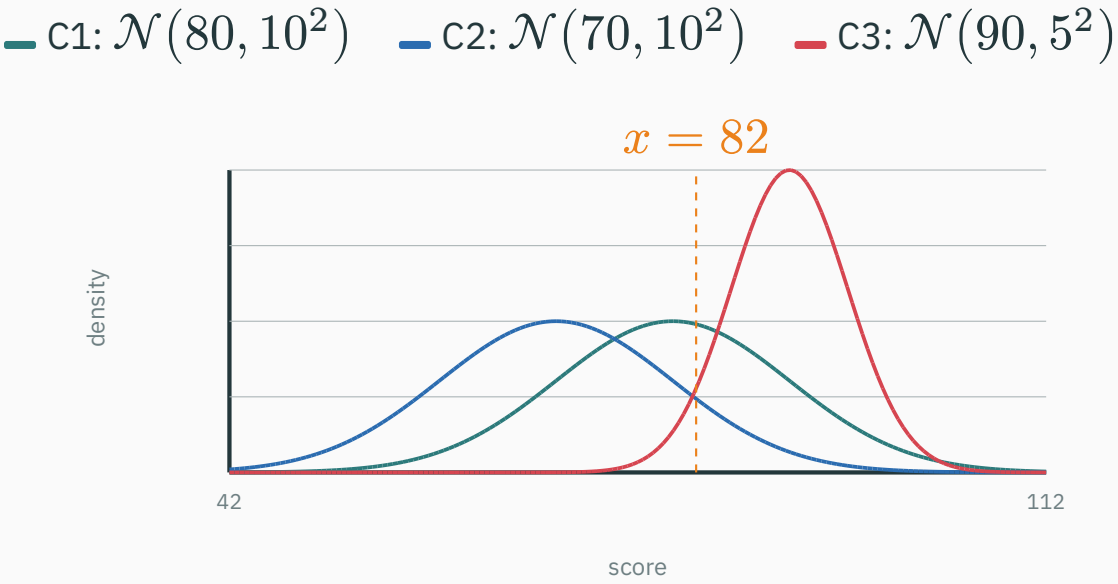
- C1: $\mathcal{N}(\mu = 80, \sigma = 10)$
- C2: $\mathcal{N}(\mu = 70, \sigma = 10)$
- C3: $\mathcal{N}(\mu = 90, \sigma = 5)$

I stop a random student in the corridor. They took exactly one of the three, and they scored **82**.

Which course would you guess? Commit to an answer — and notice what your brain just did to produce it.

The guess, made honest

V VISUAL



Read each curve **at the observed score**: $p(82 \mid \mathrm{C1}) = 0.0391$, $p(82 \mid \mathrm{C2}) = 0.0194$, $p(82 \mid \mathrm{C3}) = 0.0222$.

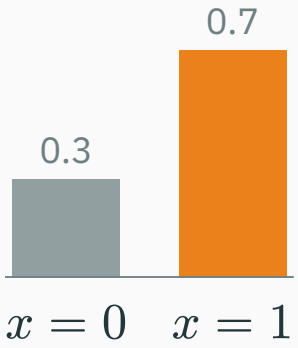
C1 wins: you picked the model under which the data is least surprising. That's maximum likelihood — you already do it by eye.

The flip, on the simplest model there is

V VISUAL

A coin with $P(\text{heads}) = \theta$. One formula, $p(x \mid \theta)$ — **two completely different readings**:

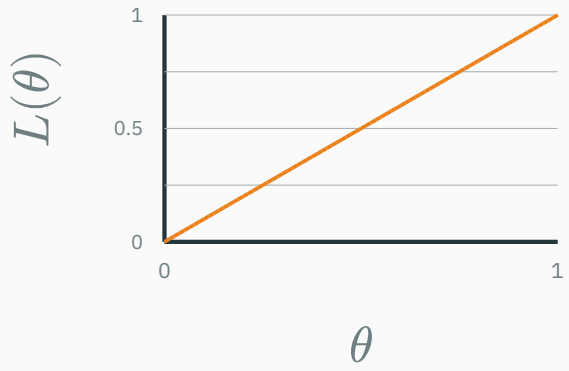
Fix $\theta = 0.7$, vary x



a **distribution** over outcomes

bars sum to 1 — guaranteed

Fix $x = 1$ (one head), vary θ



the **likelihood** of that outcome

area under this line: $1/2$ — every θ gives the same value

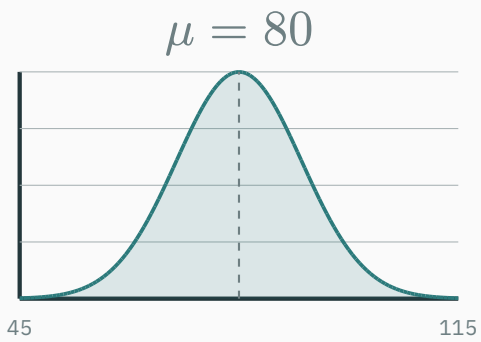
Two bars became a line: **different axis, different object** — same formula.

The flip, on the corridor student

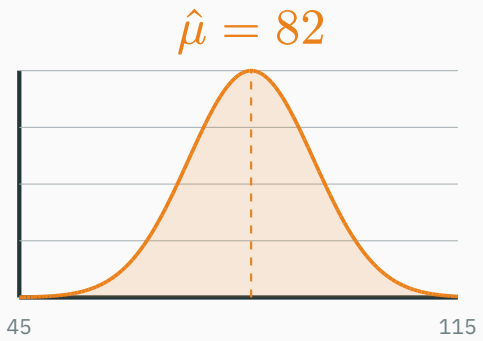
V VISUAL

Now suppose we know the course has $\sigma = 10$ but **nobody told us μ**. The student says 82.

DISTRIBUTION: fix $\mu = 80$, vary x



LIKELIHOOD: fix $x = 82$, vary μ



The right curve peaks at $\mu = 82$: the single best explanation centers the bell **on the data**. You just estimated a parameter.

Naming the move

$L(\theta) := p(\mathcal{D} \mid \theta)$, **read as a function of θ with the data $\mathcal{D}$ frozen.**

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} L(\theta).$$

- $p(x \mid \theta)$ as a function of x: a **distribution** — normalized, sums/integrates to 1
- the same formula as a function of θ: the **likelihood** — a plain score, one number per candidate model

$L(\theta)$ is **not** a probability distribution over θ: nothing makes its area 1 (the one-head line had area $1/2$). "Which θ is most probable" is a different, stronger question — that's Bayes and L15. Today we only ask which θ **explains the data best**.

The likelihood of a dataset

From one observation to a dataset

One data point gives one score: $L(\theta) = p(x_1 \mid \theta)$. A dataset $\mathcal{D} = \{x_1, ..., x_n\}$ needs the **joint** probability $p(x_1, ..., x_n \mid \theta)$ — and joints are hard. We buy simplicity with an assumption. **IID** is two separate claims:

- **Independent** — given θ, no observation tells you about another → the joint **factors into a product**
- **Identically distributed** — every observation uses the **same** $p(\cdot \mid \theta)$, one shared θ

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$$

Each claim can fail alone: tomorrow's weather depends on today's (→ L25's Markov chains); two different-noise sensors share no rule. IID is a modeling **choice** — say it out loud.

IID at work · two coins, two tiny datasets

Candidate coins: fair ($\theta = 0.5$) vs head-heavy ($\theta = 0.8$). Score both on data:

Observed	$L(0.5)$	$L(0.8)$	**Better explanation**
H, T	$0.5 \times 0.5 = 0.25$	$0.8 \times 0.2 = 0.16$	the fair coin
H, H	$0.5^2 = 0.25$	$0.8^2 = 0.64$	the head-heavy coin

Same two candidates, different frozen data → different winner. The likelihood **ranks models by the data they actually saw**.

Independence licenses the multiplication; identical distribution licenses reusing one θ in every factor.

Pop quiz, round 2 · now with a dataset

Back in the corridor. **Two** students this time — scores 82 and 72 — and both took the **same** (unknown) course. Multiply each course's two density readings:

Course	$p(82 \mid \cdot)$	$p(72 \mid \cdot)$	$L =$ **product**
C1: $\mathcal{N}(80, 10^2)$	0.0391	0.029	1.1×10^{-3}
C2: $\mathcal{N}(70, 10^2)$	0.0194	0.0391	7.6×10^{-4}
C3: $\mathcal{N}(90, 5^2)$	0.0222	0.0001	2.7×10^{-6}

C1 wins again — but look at C3: **one** far-off point (72, at $z = -3.6$) collapsed its entire product. A product is a chain of vetoes: every point must be plausible.

A dataset's likelihood multiplies one reading per point — computed with the IID recipe you just learned.

Five readings, one score

V VISUAL

Our running dataset: $\mathcal{D} = \{0.5, 1.5, 2.0, 2.5, 3.5\}$, modeled as $\mathcal{N}(\mu, 1)$ with σ known. Try $\mu = 2$: read the density at each point, multiply.

$$L(2) = 0.13 \times 0.352 \times 0.399 \times 0.352 \times 0.13 \approx 8.3 \times 10^{-4}$$

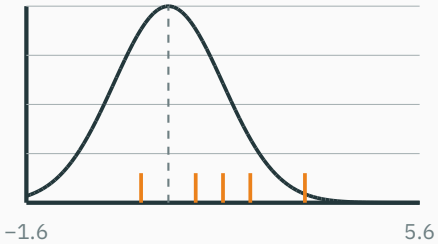
One number for the candidate $\mu = 2$. Now the key mental animation: **slide the bell and watch the score move**.

Slide the Gaussian over the data

V VISUAL

too far left

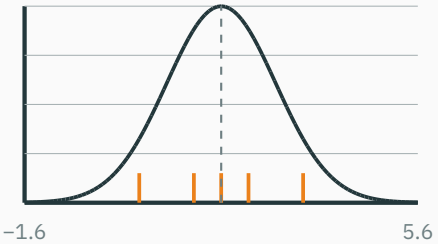
$\mu = 1$



$L(1) \approx 6.8 \times 10^{-5}$

just right

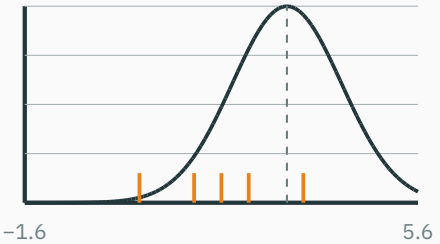
$\mu = 2$



$L(2) \approx 8.3 \times 10^{-4}$

too far right

$\mu = 3.2$



$L(3.2) \approx 2.3 \times 10^{-5}$

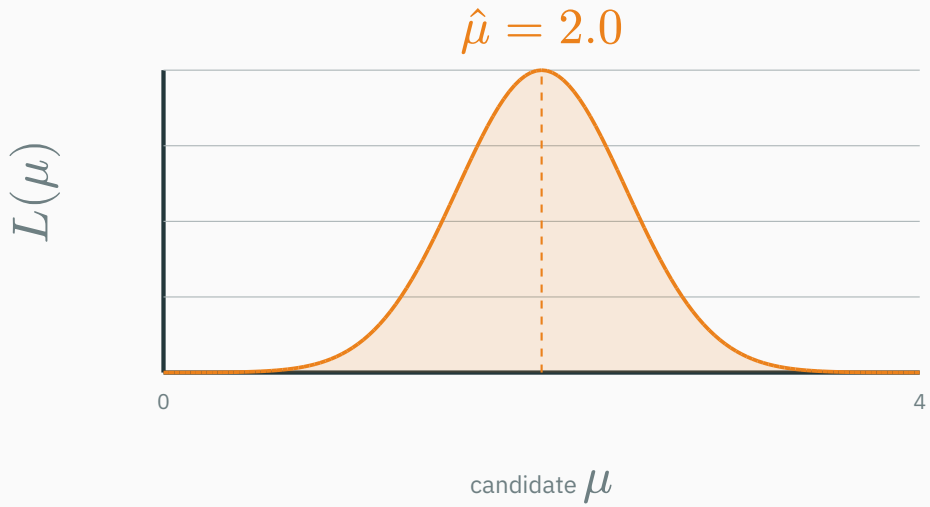
Same five orange data points every time; only the model slides. Centered wins by a factor of **12×** over $\mu = 1$.

Fitting a model = sliding its parameters until the data's likelihood is maximal.

The whole slide, as one curve

V VISUAL

Evaluate $L(\mu)$ at **every** candidate μ — the likelihood function of this dataset:



- Each point of this curve is one “slide position” from the previous slide — a **product of five densities**
- The peak sits at $\hat{\mu} = 2.0$ — which happens to be the sample mean of $\mathcal{D}$. Coincidence? We’ll **prove** it shortly.

Checkpoint: what kind of object is $L(\mu)$?

Q QUESTION

The curve $L(\mu)$ we just drew for $\mathcal{D} = \{0.5, 1.5, 2.0, 2.5, 3.5\}$ — which statement is true?

A. It is a probability density over μ, so its area must be 1

B. $L(2.0)$ is the probability that $\mu = 2.0$ is the true parameter

C. $L(\mu)$ is the joint density of the five **observed** points, re-read as a function of μ

D. Its peak location depends on the order in which the five points were multiplied

Answer: what kind of object is $L(\mu)$?

A ANSWER

Correct: C — the joint density of the observed data, as a function of μ

Why: The data is frozen; μ varies. Nothing normalizes the curve over μ (A), and no probability is ever assigned **to** μ — that requires a prior and Bayes' rule, which is L15's business (B). Multiplication is commutative, so order is irrelevant (D).

Log-likelihood: products become sums

Products of probabilities are numerically doomed

You proved this in L2 — the recap: a language model scores a 1000-character text, each character with probability ≈ 0.03:

$$P(\text{text}) = \prod_{i=1}^{1000} p_i \approx 0.03^{1000} \approx 10^{-1523} \underset{\text{even float64}}{\Rightarrow} \mathbf{0.0} \quad \text{argmax of all-zeros} = \text{garbage}$$

Our five-point likelihood already sat at 10^{-4}. In log-space, tiny becomes ordinary:

$$\log P(\text{text}) = \sum_{i=1}^{1000} \log p_i \approx 1000 \times (-3.51) = -3507 \quad \checkmark \text{ a boring, safe float}$$

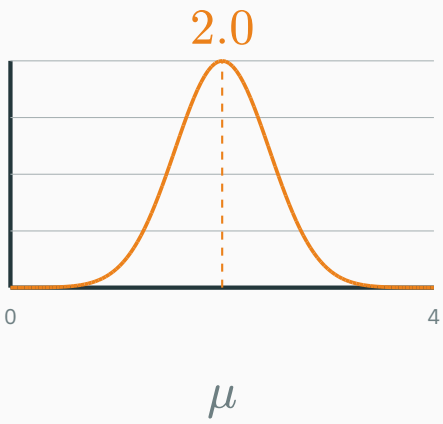
L2's exact words: "Every loss you will ever meet is a *log*-likelihood — floating point demands it (**L14 shows the deeper reason**)." Today is that day — and underflow is only reason 1 of 3.

Reason 2 · the log moves the peak nowhere

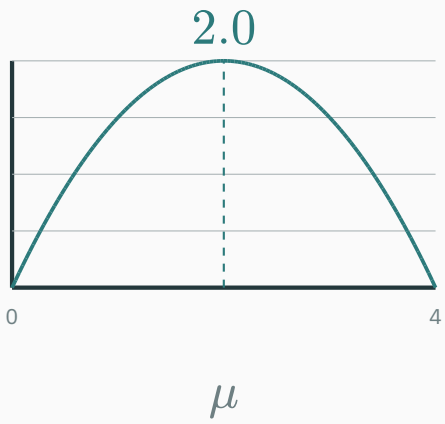
V VISUAL

$\log$ is **strictly increasing**: $a > b > 0 \Rightarrow \log a > \log b$. So it preserves every comparison of likelihoods — including where the maximum sits:

LIKELIHOOD $L(\mu)$



LOG-LIKELIHOOD $\ell(\mu)$



$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \log L(\theta)$$ — same peak, tamer numbers

Reason 3 · sums are what calculus (and autodiff) want

$$\ell(\theta) := \log L(\theta) = \log \prod_{i=1}^{n} p(x_i \mid \theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

To maximize, we differentiate (L7–L8) and set to zero. Compare the two forms:

- $\frac{\mathrm{d}}{\mathrm{d}\theta} \prod_i p_i$: the product rule over n factors — n terms, each a product of $n-1$ densities
- $\frac{\mathrm{d}}{\mathrm{d}\theta} \sum_i \log p_i = \sum_i \left(\frac{\mathrm{d}}{\mathrm{d}\theta}\right) \log p_i$: differentiate each term **alone**, add

Three independent wins — no underflow (L2), same argmax, per-point derivatives. This is why every training objective you will ever see is built from $\log p$.

Flip the sign, and you have invented “the loss”

Optimizers by convention **minimize** (Module 4 is built that way). So negate:

$$\mathrm{NLL}(\theta) = -\ell(\theta) = -\sum_{i=1}^{n} \log p(x_i \mid \theta) \qquad \text{and often the mean} \quad \frac{1}{n}\sum_{i} -\log p(x_i \mid \theta)$$

- maximize likelihood $\Leftrightarrow$ maximize log-likelihood $\Leftrightarrow$ **minimize NLL**
- each data point contributes $-\log p(x_i \mid \theta)$: **its personal penalty** — large when the model found it implausible

A “loss” is not a design flourish. Training loss = per-example negative log-likelihood, averaged. This sentence is the hinge of the whole course.

The coin: your first MLE, derived ⭐

The model, and a clever way to write it

Flip a coin 10 times: $H, H, T, H, H, H, T, H, H, T$ — **7 heads, 3 tails**. Model: each flip is Bernoulli, $P(H) = \theta$, IID. Code heads as $x = 1$, tails as $x = 0$:

$$p(x \mid \theta) = \theta^x (1-\theta)^{1-x}$$ one formula, both cases: $x = 1 \to \theta$; $x = 0 \to 1 - \theta$

The dataset's likelihood — products of θ's and $(1-\theta)$'s, one per flip:

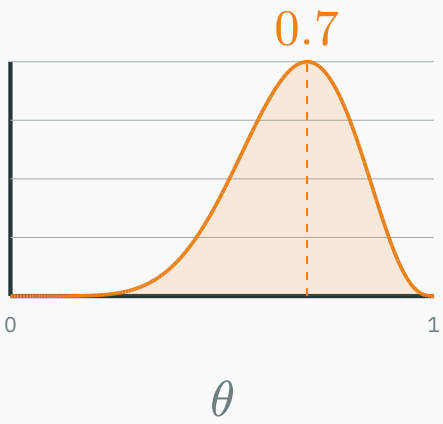
$$L(\theta) = \prod_{i=1}^{10} \theta^{x_i} (1-\theta)^{1-x_i} = \theta^7 (1-\theta)^3$$

Only the **counts** survive: 7 heads, 3 tails. The order of flips never mattered — IID told us so before we computed anything.

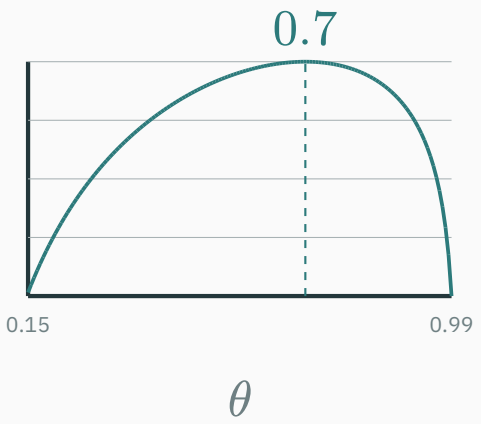
The two curves, computed

V VISUAL

$$L(\theta) = \theta^7(1-\theta)^3$$



$$\ell(\theta) = 7\log\theta + 3\log(1-\theta)$$



Both peak at $\theta = 0.7$. Spot values: $L(0.5) = 0.5^{10} \approx 9.8 \times 10^{-4}$, $L(0.7) \approx 2.2 \times 10^{-3}$ — the data is 2.3× more plausible under $\theta = 0.7$ than under a fair coin.

⭐ Derivation · set the derivative to zero

D DERIVATION

Maximize $\ell(\theta) = 7\log\theta + 3\log(1-\theta)$ the L7 way — stationary point:

$$\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \frac{7}{\theta} - \frac{3}{1-\theta} = 0$$

$$7(1-\theta) = 3\theta \quad\Rightarrow\quad 7 = 10\theta \quad\Rightarrow\quad \hat{\theta} = \frac{7}{10} = 0.7$$

Is it a maximum? Check curvature (L7's second-derivative test):

$$\frac{\mathrm{d}^2\ell}{\mathrm{d}\theta^2} = -\frac{7}{\theta^2} - \frac{3}{(1-\theta)^2} < 0 \quad \text{everywhere on } (0,1)$$

Strictly concave → the single stationary point is the **global** maximum. No luck involved.

⭐ Derivation · now for any dataset

General case: n flips, n_H heads, $n_T = n - n_H$ tails. Same three moves:

$$\ell(\theta) = \sum_{i=1}^{n} [x_i \log\theta + (1 - x_i)\log(1-\theta)] = n_H \log\theta + n_T \log(1-\theta)$$

$$\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \frac{n_H}{\theta} - \frac{n_T}{1-\theta} = 0 \quad \Rightarrow \quad n_H(1-\theta) = n_T\theta$$

$$n_H = (n_H + n_T)\theta$$

$$\hat{\theta}_{\mathrm{MLE}} = \frac{n_H}{n} \textbf{ — the fraction of heads.}$$

The answer your intuition would have guessed — but now it is a **theorem**, from a principle that will still work when intuition has nothing to say (a billion-parameter model has no “fraction of heads”).

Sanity checks, one honest and one alarming

The machine agrees — brute-force the curve:

```
heads, n = 7, 10
theta  = np.linspace(0.01, 0.99, 981)
loglik = heads * np.log(theta) + (n - heads) * np.log(1 - theta)
theta[np.argmax(loglik)]                    # 0.7  — the head fraction
```

Now flip a coin **3 times**, get **3 heads**: $\hat{\theta}_{\text{MLE}} = 3/3 = 1.0$ — this coin **cannot ever** land tails. Three flips convinced MLE of a certainty. Your intuition says “probably just a normal coin”. Your intuition is using a **prior** — and giving MLE one is exactly L15.

Interactive · climb the coin's likelihood yourself

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mle-map-coin

The **mle-map-coin** article flips coins live: the likelihood curve rebuilds after every flip, and you drag θ to feel the score change under your cursor. Recreate today's 7-of-10, then try 3-of-3 and watch the peak slam into the boundary at $\theta = 1$. (Ignore the prior slider for now — after L15 it will be the whole point.)

Two-minute experiment before next lecture: keep the head **fraction** at 70% but grow n: 7/10 → 70/100 → 700/1000. Watch what happens to the curve's **width** — we name that phenomenon at the end of today.

MLE for the Gaussian

The Gaussian log-likelihood, simplified once and for all

Model $\mathcal{D} = \{x_1, \ldots, x_n\}$ as IID $\mathcal{N}(\mu, \sigma^2)$ (L12's density, L13's star):

$$\ell(\mu, \sigma) = \sum_{i=1}^{n} \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \right]$$

$\log$ of (product of exp) — everything unfolds:

$$\ell(\mu, \sigma) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

All the μ-dependence sits in one term: $-\sum_i (x_i - \mu)^2$. Maximizing ℓ over μ is minimizing a sum of squares. Remember this shape — it returns in ten minutes as MSE.

The mean, earned

Differentiate w.r.t. μ (the constants and $-n \log \sigma$ vanish):

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0 \quad \Rightarrow \quad \sum_i x_i = n\mu$$

$$\hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x}$$

On our running dataset: $\hat{\mu} = (0.5 + 1.5 + 2.0 + 2.5 + 3.5)/5 = 2.0$ — **exactly where the likelihood curve peaked** in Section 3. The picture and the calculus agree.

Note what $\hat{\mu}$ does **not** depend on: σ. However noisy the ruler, the best center is the average. (With σ unknown, the μ-equation is unchanged.)

The variance — and an honest footnote

Same recipe for σ: differentiate $\ell = \ldots - n\log\sigma - \frac{1}{2\sigma^2}\sum_i (x_i-\mu)^2$, set to zero, solve:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_i (x_i - \hat{\mu})^2 = 0 \quad \Rightarrow \quad \hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

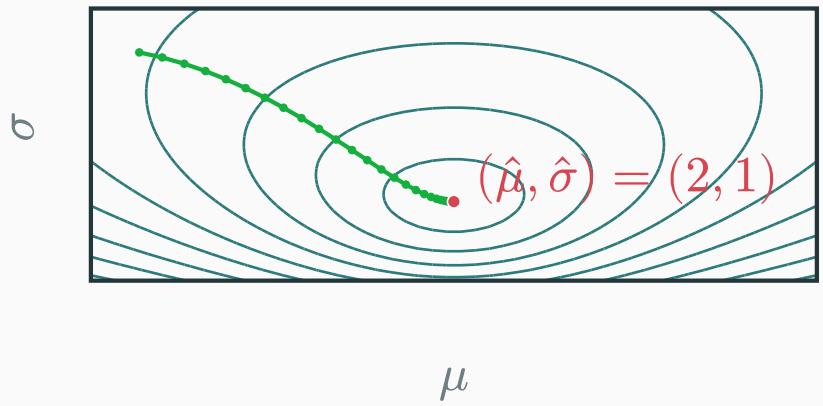
On $\mathcal{D}$: deviations $(-1.5, -0.5, 0, 0.5, 1.5)$, squares sum to 5.0, so $\hat{\sigma}^2 = 5.0/5 = 1.0$.

Honest footnote. This estimator is **biased**: on average across datasets, $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\cdot\sigma^2$ — systematically a bit small, because $\bar{x}$ chased the sample before we measured spread around it. NumPy's `np.var(x, ddof=1)` applies the $\frac{n}{n-1}$ repair (Bessel's correction). We state this; a statistics course proves it. MLE is **excellent**, not sacred.

Estimation is optimization · the NLL landscape

V VISUAL

Free both parameters: the **dataset NLL** over the (μ, σ) plane, computed live from $\mathcal{D}$ —



- The bowl bottoms out at $(\hat{\mu}, \hat{\sigma}) = (2.0, 1.0)$ — both closed forms at once
- The green path: **gradient descent** walking downhill on this exact surface

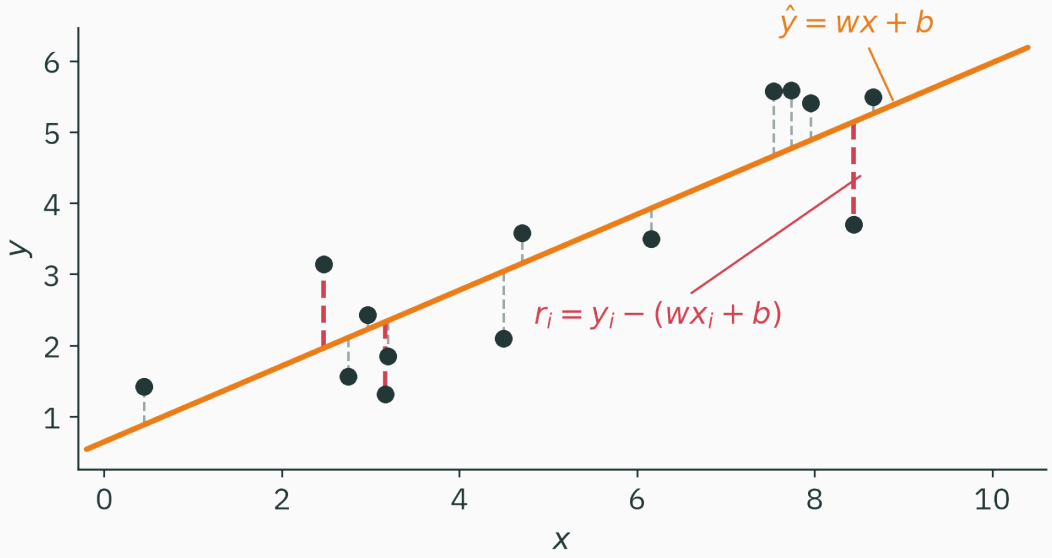
Fitting = minimizing NLL. Module 4 (L16–L21) is the art of walking downhill on such surfaces.

The keystone: linear regression → MSE

A model that finally has an input

V VISUAL

Everything so far fit a **constant** distribution. Real problems predict y **from** x — rent from area, temperature from hour. The simplest such model is a line:



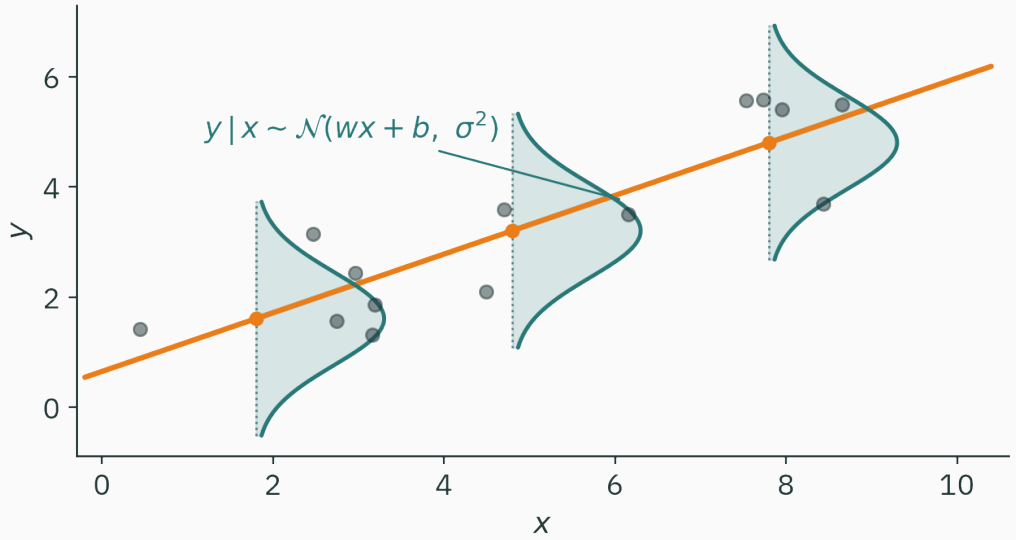
$\hat{y} = wx + b$; the **residual** $r_i = y_i - (wx_i + b)$ is what the line cannot explain. Which line is **best**? We need a likelihood — so we need a **distribution over** y.

The noise model: a bell standing on the line

V VISUAL

Assume the world draws y as **the line plus Gaussian noise**:

$$y_i = wx_i + b + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ IID} \quad \Rightarrow \quad y_i \mid x_i \sim \mathcal{N}(wx_i + b, \sigma^2)$$



A vertical Gaussian stands **on the line** at every x: near the line, high density; stragglers, expensive. That is the whole assumption — now we turn the crank.

The likelihood of a **labeled** dataset

New situation, same machinery. The data comes in pairs (x_i, y_i), and our model only says how y arises **given** x — it never models the inputs:

$$L(\theta) = \prod_{i=1}^{n} p(y_i \mid x_i, \theta) \qquad \theta = (w, b);\ \text{the } x_i \text{ enter as fixed givens}$$

- Same flip: data (x_i, y_i) frozen, parameters (w, b) varying
- Same IID product — independence **of the noise** ε_i, one shared line
- No $p(x)$ anywhere: whoever chose the inputs is not our problem

This conditional form is the shape of **every** supervised objective — including the trillion-token ones: L26's language model maximizes exactly such a product, one conditional per next character.

⭐ Derivation · one point's log-likelihood

D DERIVATION

The density of observing label y_i at input x_i, given parameters $\theta = (w, b)$:

$$p(y_i \mid x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (wx_i + b))^2}{2\sigma^2} \right)$$

Take the log — the same unfold as the Gaussian section:

$$\log p(y_i \mid x_i, \theta) = -\frac{(y_i - (wx_i + b))^2}{2\sigma^2} - \log\sigma - \frac{1}{2}\log(2\pi)$$

$$-\log p(y_i \mid x_i, \theta) = \frac{\underbrace{(y_i - \hat{y}_i)^2}_{\text{squared residual!}}}{2\sigma^2} + \underbrace{\log\sigma + \frac{1}{2}\log(2\pi)}_{\text{no } \theta \text{ inside — constant}}$$

One data point's NLL **is** its squared residual, rescaled and shifted by constants.

⭐ Derivation · sum, strip constants, done

D DERIVATION

Sum over the IID dataset and maximize over $\theta = (w, b)$, with σ fixed:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ell(\theta) = \arg\min_{\theta} \sum_{i=1}^{n} \left[\frac{(y_i - \hat{y}_i)^2}{2\sigma^2} + \log\sigma + \frac{1}{2}\log(2\pi) \right]$$

Adding a constant never moves an argmin — drop $n \log\sigma + \frac{n}{2}\log(2\pi)$. Scaling by a positive constant never moves an argmin — drop $1/(2\sigma^2)$:

$$\hat{\theta}_{\text{MLE}} = \arg\min_{w,b} \sum_{i=1}^{n} (y_i - wx_i - b)^2$$ **— Gaussian MLE is least squares.**

Nobody "invented" MSE: it was hiding inside $\mathcal{N}$ all along.

What MSE silently assumes

Run the derivation backwards: using MSE **is** declaring a noise model. Three claims, made every time you call `mse_loss`:

1. noise is **Gaussian** — bell-shaped errors around the prediction
2. noise has **the same σ everywhere** — no input is harder than another
3. observations are **independent** given the inputs

Assumption 1 is how outliers wreck least squares: a point 5σ off the line pays $(5\sigma)^2/2\sigma^2 = 12.5$ nats — the Gaussian is **certain** such points are freakish, so the fitted line contorts to appease them. Heavier-tailed noise models forgive them — that's the MAE row coming up.

The machine agrees · one grid, two objectives

```
x  = np.array([0., 1., 2., 3., 4.]); y = np.array([1.1, 1.9, 2.8, 4.3, 4.9])
ws = np.linspace(0.2, 1.8, 801)                 # candidate slopes (b = 1 fixed)
mse = [np.mean((y - w*x - 1.0)**2) for w in ws]
ll  = [np.sum(-0.5*np.log(2*np.pi) - 0.5*(y - w*x - 1.0)**2) for w in ws]
print(ws[np.argmin(mse)], ws[np.argmax(ll)])    # 1.0 1.0  — the same slope
```

Two objectives, one winner — on every dataset, always, by the derivation you just did.

This is why earlier stats courses could teach least squares without ever saying "likelihood": with σ fixed, the two problems are **literally the same problem**. You now know which one is fundamental.

Every loss is an NLL

The table this course orbits around

Choose a noise/observation model; the NLL **is** your loss:

You observe	Model	NLL per example	Its famous name
a real y	Gaussian	$(y-\hat{y})^2/2\sigma^2 + c$	**MSE** ✓ today
yes / no	Bernoulli	$-[y \log p + (1-y)\log(1-p)]$	**BCE** ✓ today
1 of K classes	Categorical	$-\log p_{\text{true class}}$	**cross-entropy** → L24
y with outliers	Laplace	$\lvert y-\hat{y}\rvert/b + c$	**MAE** bonus

Loss functions are not a menu of arbitrary formulas. Each row is the same construction — only the noise model changes.

The BCE row costs one line

For a yes/no label $y \in \{0, 1\}$ with predicted head-probability p, the Bernoulli pmf in exponent form (from the coin section!) gives the per-example NLL:

$$-\log\left[p^{y}(1-p)^{1-y}\right] = -\left[y \log p + (1-y)\log(1-p)\right]$$

This is `binary_cross_entropy`, derived directly from the Bernoulli log-likelihood.

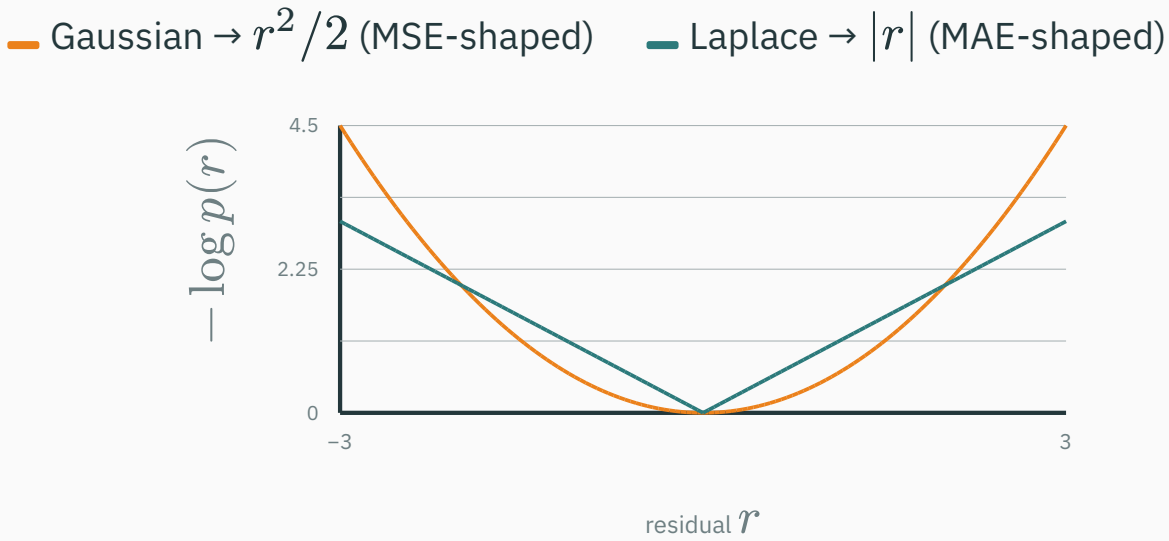
- $y = 1$, model said $p = 0.9$: penalty $-\log 0.9 \approx 0.105$ — cheap
- $y = 1$, model said $p = 0.01$: penalty $-\log 0.01 \approx 4.6$ — confident wrongness is **expensive**, by the shape of $-\log$

The categorical / cross-entropy row is the same move over K classes — L24 does it properly and connects it to information theory (why "entropy" is in the name), and L2′s stable log-softmax is exactly how it's computed.

Two noise models, two loss shapes

V VISUAL

The **true** per-residual loss curves, computed from the densities themselves ($-\log p$, shifted to 0 at $r = 0$):



The parabola **accelerates**: doubling a large residual quadruples its cost — outliers steer the fit. The Laplace's thin tails in log-space charge a flat rate — outliers get a shrug. Choosing a loss **is** choosing how much you believe in outliers.

Match each common loss to an observation model

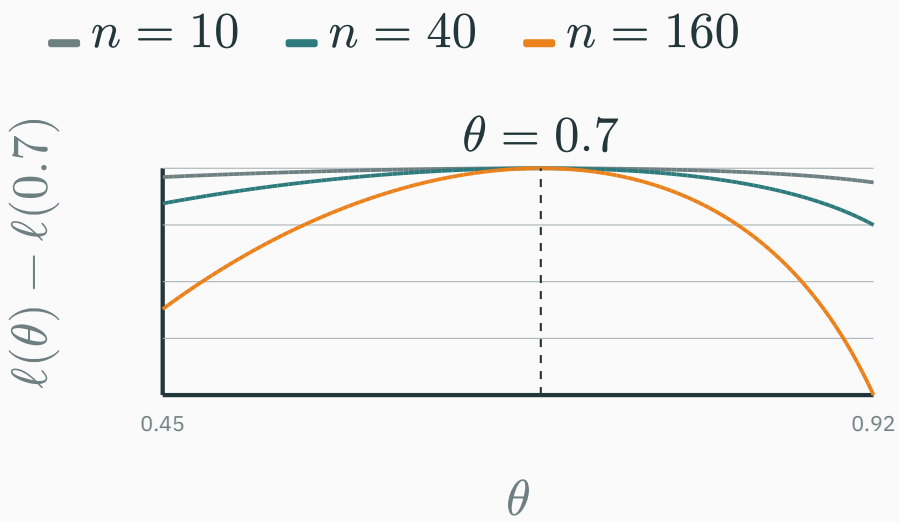
Loss	Where it comes from	Status
`mse_loss`	Gaussian noise on the prediction — derived, boxed, yours	**closed**
`binary_cross_entropy`	Bernoulli NLL — the coin, renamed	**closed**
`cross_entropy`	Categorical NLL + information theory: **why it's called entropy**, what it has to do with compression	**open → L24**

L2 told you `CrossEntropyLoss` fuses log-softmax for stability and takes raw logits. Today you learned why it's a **loss** at all. L24 explains why it's called **entropy**. Three lectures, one function — that's how central it is.

Properties, in brief · more data sharpens the peak

V VISUAL

Hold the head fraction at 70%, grow n — the peak-aligned log-likelihood sharpens:



- **Consistency** (stated): as $n \to \infty$, $\hat{\theta}_{\text{MLE}} \to$ the true θ — alternatives get buried
- Small n misbehaves: biased $\hat{\sigma}^2$, certainty from 3 flips. Data cures MLE; priors (L15) cure small data.

Checkpoint: choose the loss like a modeler

Q QUESTION

Your sensor's errors are usually small but show occasional huge spikes — well modeled by Laplace noise. Maximum likelihood hands you which training loss?

A. MSE — squared error is always optimal

B. MAE — the mean **absolute** error

C. BCE — it has a log, logs handle spikes

D. Cross-entropy over discretized bins

Answer: choose the loss like a modeler

A ANSWER

Correct: B — MAE — mean absolute error

Why: Laplace density $\propto e^{-|r|/b}$, so per-point NLL $= |r|/b + c$: the absolute residual. The quadratic MSE would let each spike bully the fit (r^2 explodes); the linear penalty stays calm. Same construction as MSE — different noise belief. That is the table doing real work.

Practice problems

Try on paper; the T7 notebook (after L15) checks all of them in NumPy.

1. 12 flips, 9 heads: write $\ell(\theta)$, maximize it, and compute $\ell(0.75) - \ell(0.5)$.
2. For $\mathcal{D} = \{1.0, 3.0\}$ under $\mathcal{N}(\mu, 1)$: sketch $L(\mu)$; find $\hat{\mu}$, then $\hat{\sigma}^2$. (Answers: 2.0 and 1.0.)
3. Exponential waiting times $p(x \mid \lambda) = \lambda e^{-\lambda x}$ (L12's zoo): derive $\hat{\lambda}_{\text{MLE}} = 1/\bar{x}$.
4. Redo the keystone with **Laplace** noise and show the objective becomes $\sum_i |y_i - wx_i - b|$.
5. For $n = 10^6$, $p_i \approx 0.1$: estimate $\prod_i p_i$ and compare with float64's smallest positive number ($\approx 10^{-324}$, L2). Why must training code never form this product?
6. True or false, one sentence: "the MLE of θ is the most probable value of θ given the data." Explain the distinction introduced in L15.

Lecture 14 — summary

- **The flip**: freeze the data, vary θ — the likelihood is a score, not a distribution over θ.
- **Datasets**: IID → a product of per-point densities; slide the model, climb the score.
- **Logs**: same argmax, no underflow (L2′s 0.03^{1000}, cashed), per-point derivatives. NLL = **the loss**.
- **Coin** ⭐: $\hat{\theta} = n_H/n$ from $\ell'(\theta) = 0$; 3-for-3 heads exposes MLE's blind spot (→ L15).
- **Gaussian**: $\hat{\mu} = \bar{x}$; $\hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$ — biased, honestly footnoted.
- **Keystone** ⭐: Gaussian noise on a line → max likelihood ≡ min squared error.
- **The table**: Gaussian→MSE, Bernoulli→BCE, Categorical→CE (L24), Laplace→MAE.

Read before L15 — MML §8.1–8.3 (skim 8.1–8.2, focus 8.3) · MacKay Ch. 2. **T7** (after L15) drills MLE + MAP together.

⭐⭐⭐ Where MLE points as $n \to \infty$

★ OPTIONAL

Divide the log-likelihood by n: an **average** that (by large numbers) approaches an expectation over the true data source $p^\star$:

$$\frac{1}{n}\ell(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log p(x_i \mid \theta) \quad \underset{n\to\infty}{\to} \quad \mathbb{E}_{x\sim p^\star}[\log p(x \mid \theta)]$$

Add and subtract the entropy of $p^\star$ (L22 defines it properly):

$$\mathbb{E}_{p^\star}[\log p(x \mid \theta)] = \underbrace{\mathbb{E}_{p^\star}[\log p^\star(x)]}_{\text{fixed: no } \theta} - \underbrace{\mathrm{KL}(p^\star \,\|\, p_\theta)}_{>= 0, \text{ gap to reality}}$$

Maximizing likelihood is equivalent to minimizing $\mathrm{KL}(p^\star \,\|\, p_\theta)$ up to a constant independent of θ. L24 derives this identity from cross-entropy.

Every loss function is a negative log-likelihood in disguise.

Next: **MAP & Conjugate Priors** — what if you have beliefs *before* seeing data?